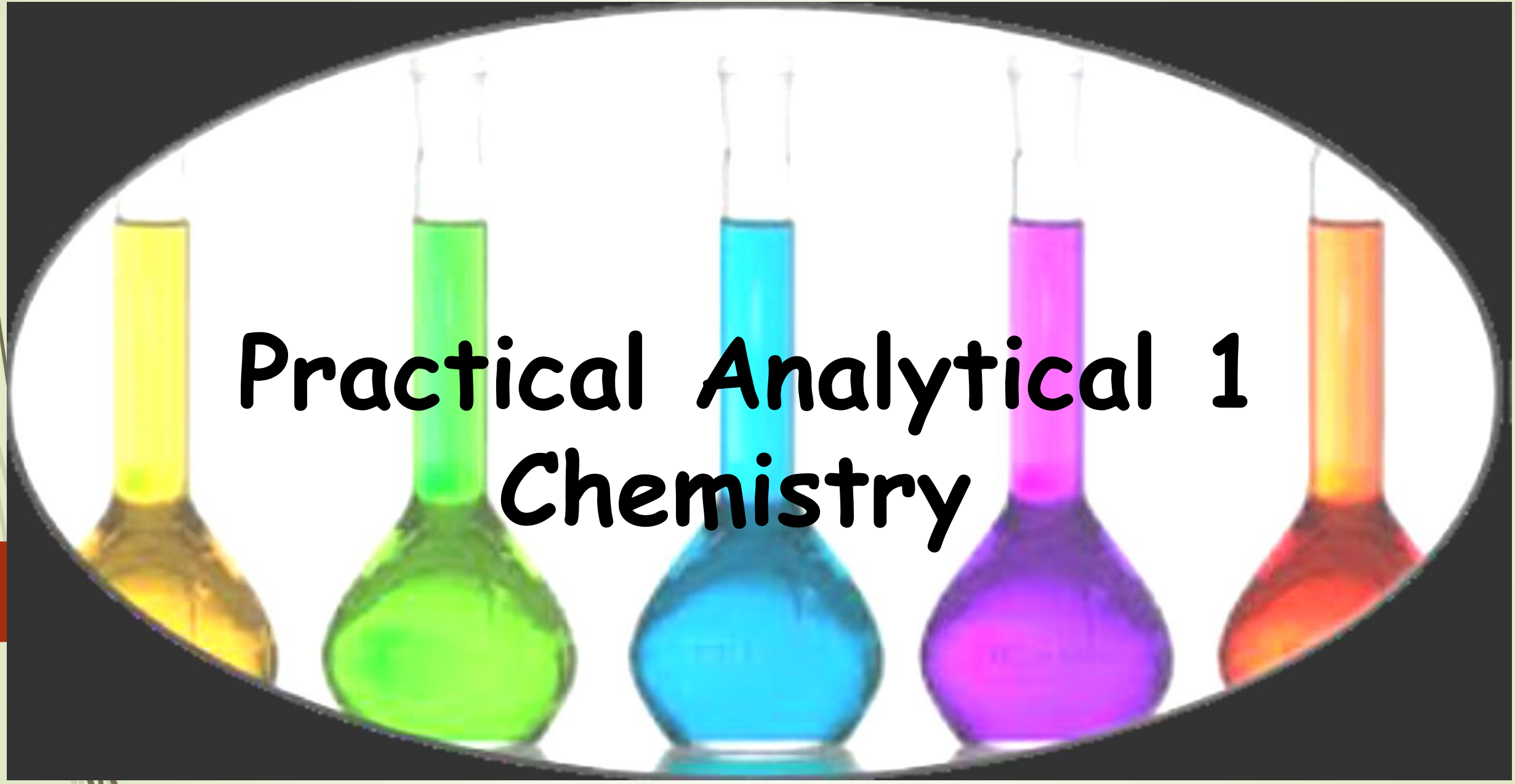


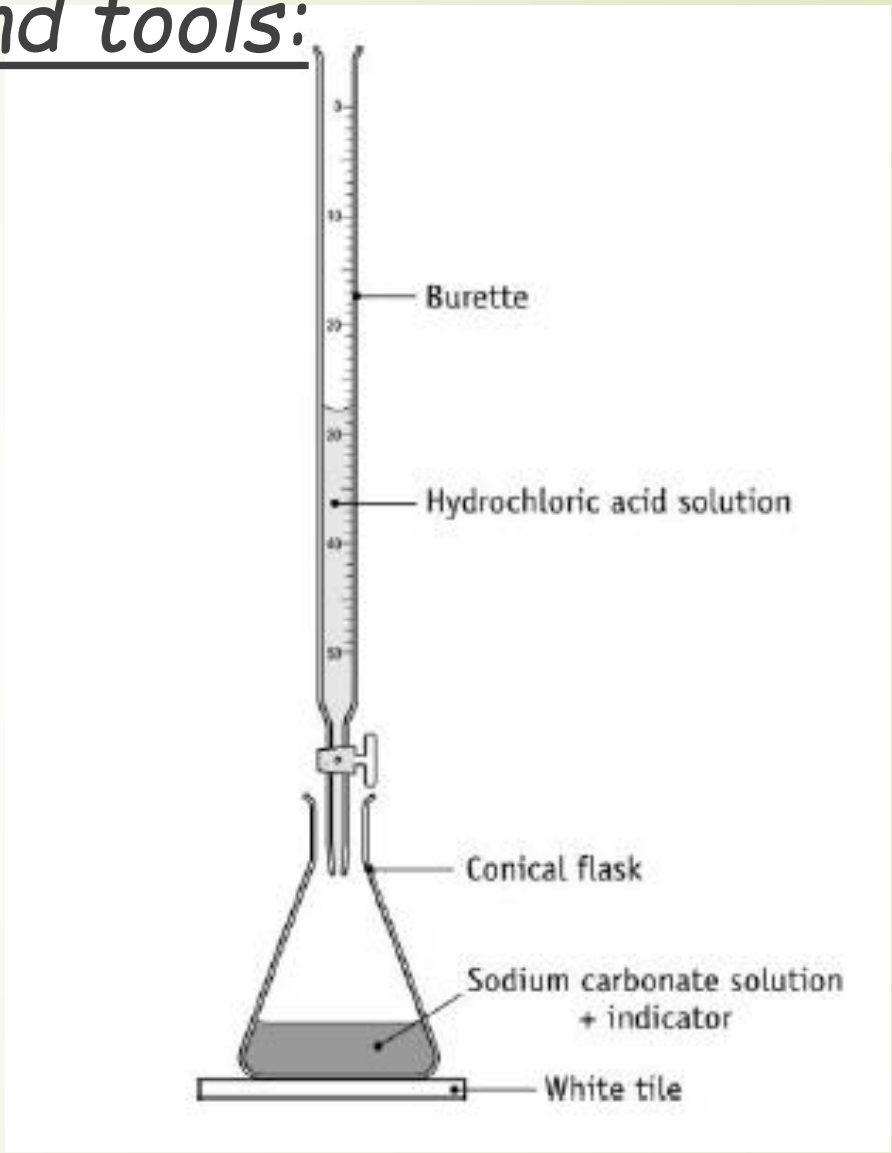
Practical Analytical 1 Chemistry

Experimental 2:

Standardization of HCL using standard solution of Na_2CO_3

Chemicals and tools:

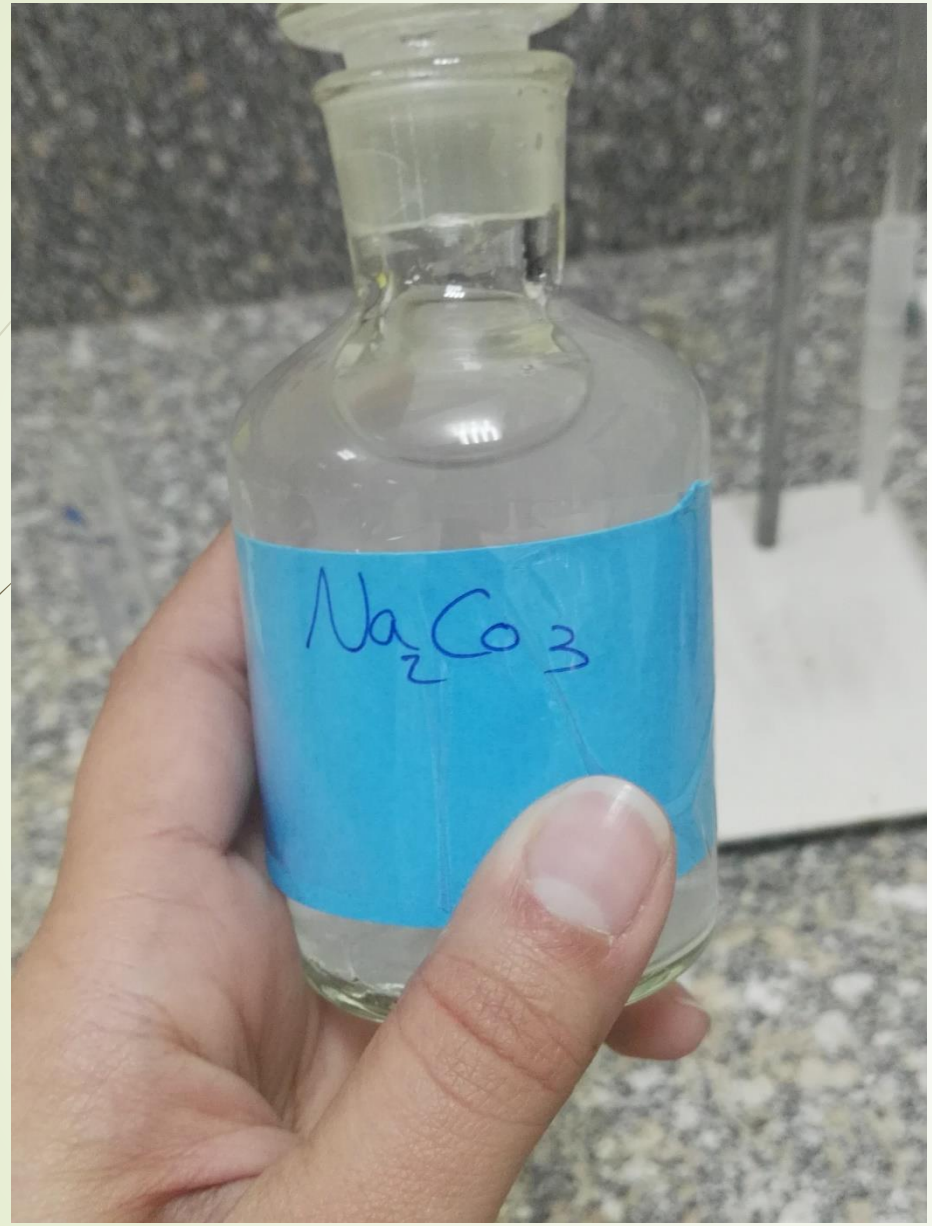
- 25-mL Burette
- 250-mL conical flasks
- Burette funnel
- 10-mL pipette
- Graduated Cylinder
- HCL(aq)
- Na_2CO_3(aq)
- Methyl Orange (M.O.)
- Distilled water



Procedure:

- Add the acid solution to the burette.
- Draw 10 mL of the base solution into the volumetric pipette and transfer this solution into a conical flask.
- Add 1-2 drops of M.O to the base solution in the flask.
- Place the flask under the burette and start adding the acid solution to the conical flask, until the color change from yellow to orange.
- Record the final reading of the burette. Wash the contents of the flask down the drain with water.
- Calculate the concentration of Na_2CO_3.

HCL


Na2Co3

Calculation

- Suppose that the volume of HCL is V1 and its normality is N1 while V2 is the volume taken from Na_2CO_3 and N2 is its normality.
- The volume of HCL (from burette) = V1
- **$(N1 \times V1)_{HCL} = (N2 \times V2)_{Na2CO3}$**
- **$(0.1 \times V1_{exp})_{HCL} = (N2 \times 10)_{Na2CO3}$**
- **$0.1 \times 8.7 = N2_{Na2CO3} \times 10$**
- **$N2_{Na2CO3} = 0.09$ N**
- Concentration in gm/L = normality x eq. wt.
- **Concentration of Na_2CO_3 = N1 x 40= 0.09 x 40= 3.5 g/L**

Problems

- 30. mL of a HBr solution is titrated with 0.075 M KOH. If 39.1 mL of KOH is required to reach the equivalence point, what is the concentration of the acid?
- 45.0 mL of a NaOH solution is titrated with 0.12 M HCL. If 29.6 mL is required for neutralization, what is the concentration of the NaOH solution?
- Show colors of CH_3COONH_4 in M.O. and Ph.Ph?
- Calculate number of moles if (0.4N) of H_3PO_4 dissolved in 500ml Distilled water?
- Calculate Equivalent moles if (0.4N) of Na_3PO_4 dissolved in 500ml Distilled water?

Problems

- *Calculate Equivalent weight:*

  H_3PO_4

  $NaHCO_3$

  $(NH_4)_2CO_3$

  $ALCL_3$

- *Calculate PPm*:

  (0.3 gm/ml) of H_3PO_4

  (5 gm/L) of $NaHCO_3$

  (0.6N) of $(NH_4)_2CO_3$

  (0.5M) of $ALCL_3$

  (H=1, Na=23, P=31, O=16, C=12, N=14, AL=27, CL=35.5)